Rocky Balboa Project

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March 2023

1 Abstract

The project is a classic problem from control theory-to balance an inverted pendulum on a horizontally-translating cart. This is one of the most commonly used representations for an unstable system. We will be using a modified Balboa (hence the double-pun nickname, 'Rocky') 32U4 Balancing Robot Kit from Pololu Robotics and Electronics. Our job is to design control systems that keep the robot upright when it's subject to an impact disturbance.

2 Initial System

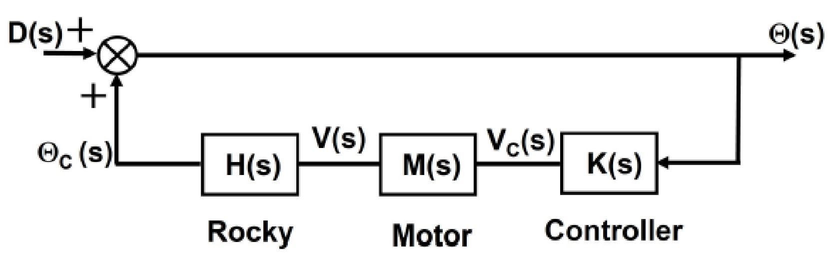

Figure 1: Block diagram of the initial system.

The project starts with the simulation-based design of an initial system in which the control system maintains balance by keeping its body (pendulum) in the vertical position and rejects any angular disturbance. The transfer function for Rocky to be $H(s) = \frac{\theta_c(s)}{V(s)}$ and used a PI controller, $K(s) = K_p + \frac{K_i}{s}$. We assumed that the motor model can be represented as a first order system where its transfer function

$$M(s) = \frac{V(s)}{V_c(s)} = \frac{\frac{a}{b}}{s + \frac{1}{b}} = \frac{\frac{K}{\tau}}{(s + \frac{1}{\tau})} \tag{1}$$

relates the velocity of the cart (output) to the motor (velocity) control signal, $V_c(s)$ (input).

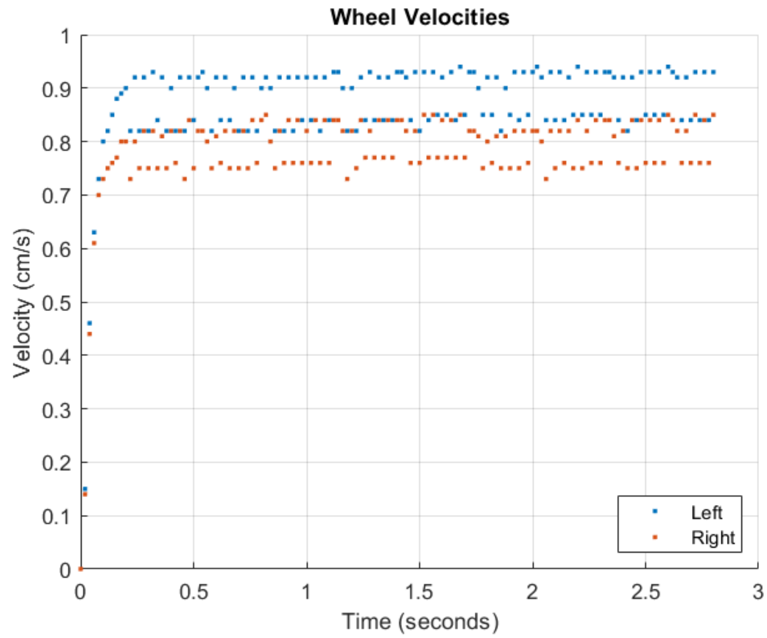

Figure 2: Wheel velocities of the left and right wheels respectively.

2.1 Parameter Extraction

The given motor transfer function is:

$$\frac{V_{out}}{V_c} = \frac{\frac{K}{\tau}}{(s + \frac{1}{\tau})}, \tag{2}$$

where K is a motor constant and τ is the time constant for the system. For the step input

$$V_c = \frac{V_c}{s}, \tag{3}$$

the response in the s domain is

$$V_{out} = \frac{\frac{K}{\tau}}{s(s + \frac{1}{\tau})}. \tag{4}$$

Using laplace transform tables, we get the time response:

$$V_{out}(t) = Kv_c - Kv_c \cdot e^{\frac{-1}{\tau}t}. \tag{5}$$

We used Matlab curve fitting tool to fit a curve of the form,

$$a - a \cdot e^{\frac{-t}{b}}, \tag{6}$$

to the data, from which we obtained the values of τ and K. When we fit the parameters in the equation a was equivalent to Kv_c and b was equivalent to τ

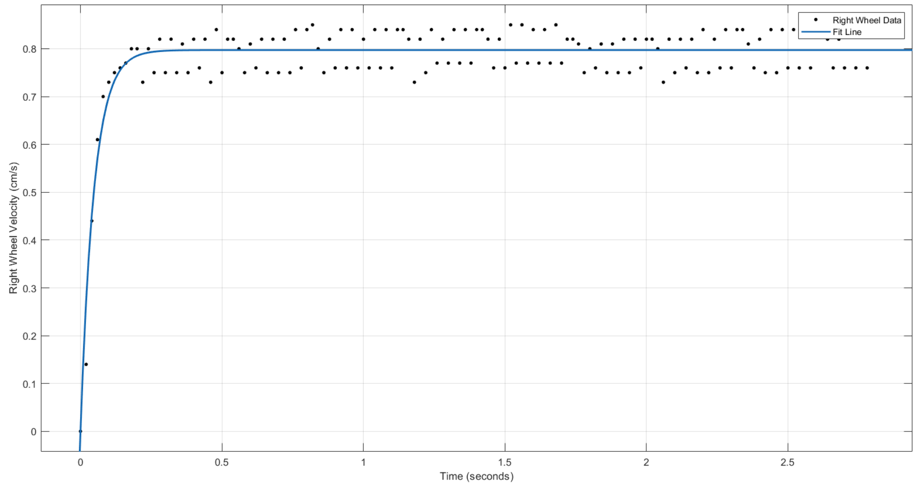

Figure 3: Fit line to the right wheel velocity for parameter extraction.

```
General model:
    f(x) = a-a*exp(-(1/b)*x)
Coefficients (with 95% confidence bounds):
     a =       0.886  (0.8782, 0.8937)
     b =     0.05128  (0.04471, 0.05786)
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Figure 4: Extracted values for a and b from the right wheel velocity.

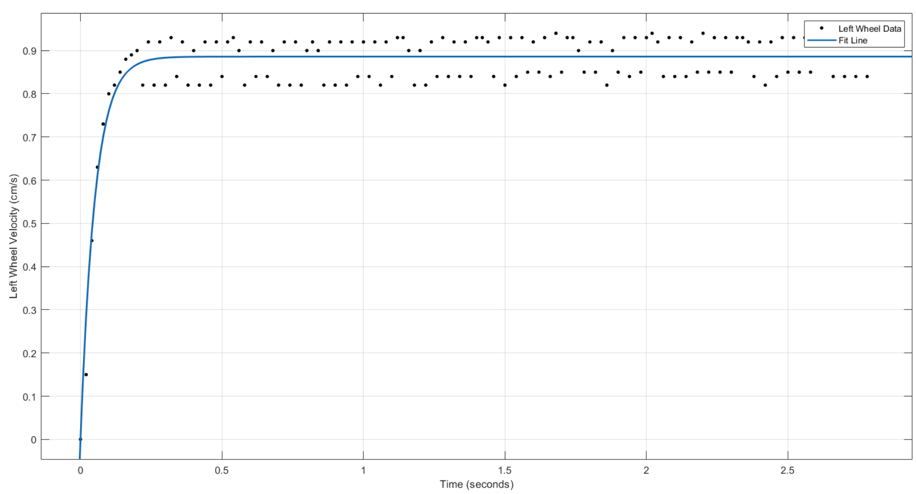

Figure 5: Fit line to the left wheel velocity for parameter extraction.

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General model:
    f(x) = a-a*exp(-(1/b)*x)
Coefficients (with 95% confidence bounds):
      a =      0.7973  (0.7906, 0.8039)
      b =     0.04767  (0.04162, 0.05373)
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Figure 6: Extracted values for a and b from the right wheel velocity.

We then found needed to find the natural frequency of the system.

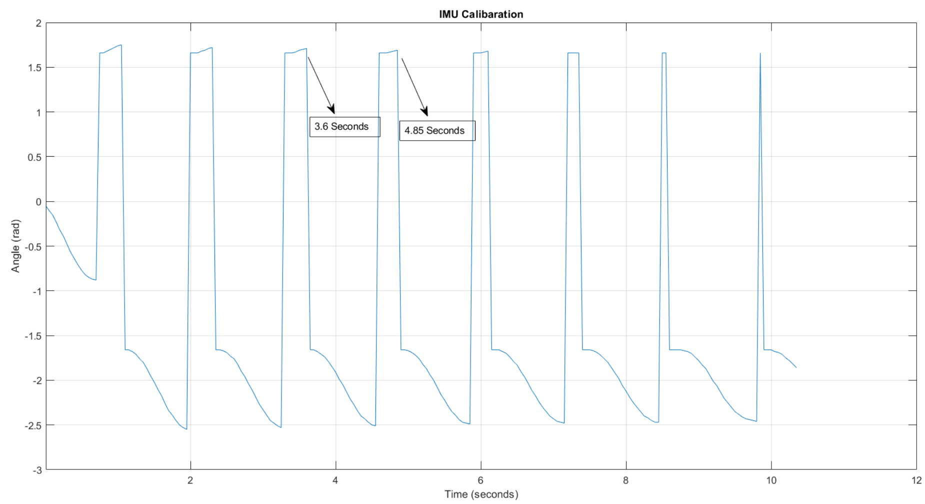

Figure 7: Calibration data to determine the natural frequency of the system.

To determine the natural frequency, we hung the robot by the wheel axles and collected accelerometer data. We then determined the time between peaks to get the period, from which we obtained the natural frequency. Based on the plot shown above, we got a frequency of

$$\omega_n = 4.65421 rad/s. \tag{7}$$

To find the effective length, we used the equation $l_{eff} = \frac{g}{\omega_n^2}$, $(g = 9.8)$, which is derived from the definition of natural frequency. We got an effective length of $l = 0.4524$.

Kv_{cL}	τ_L	Kv_{cR}	τ_R	ω_n
0.886	0.05128	0.7973	0.04767	$4.65421 rad/s$

2.2 Response of the Initial System

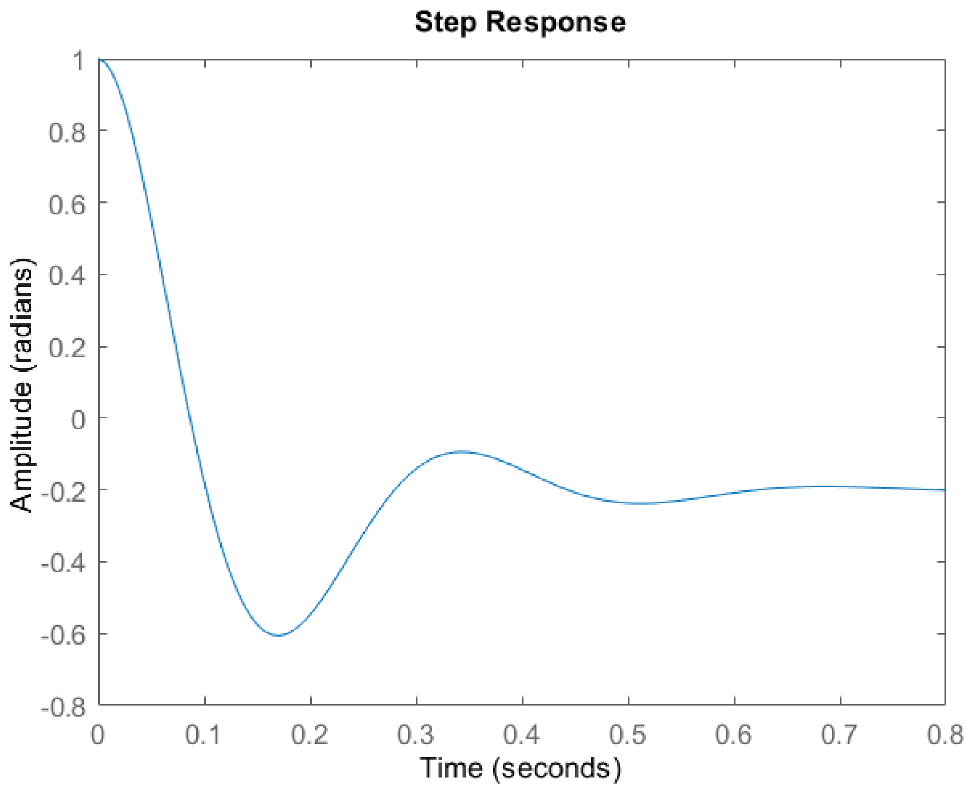

Figure 8: Step Response of the initial system.

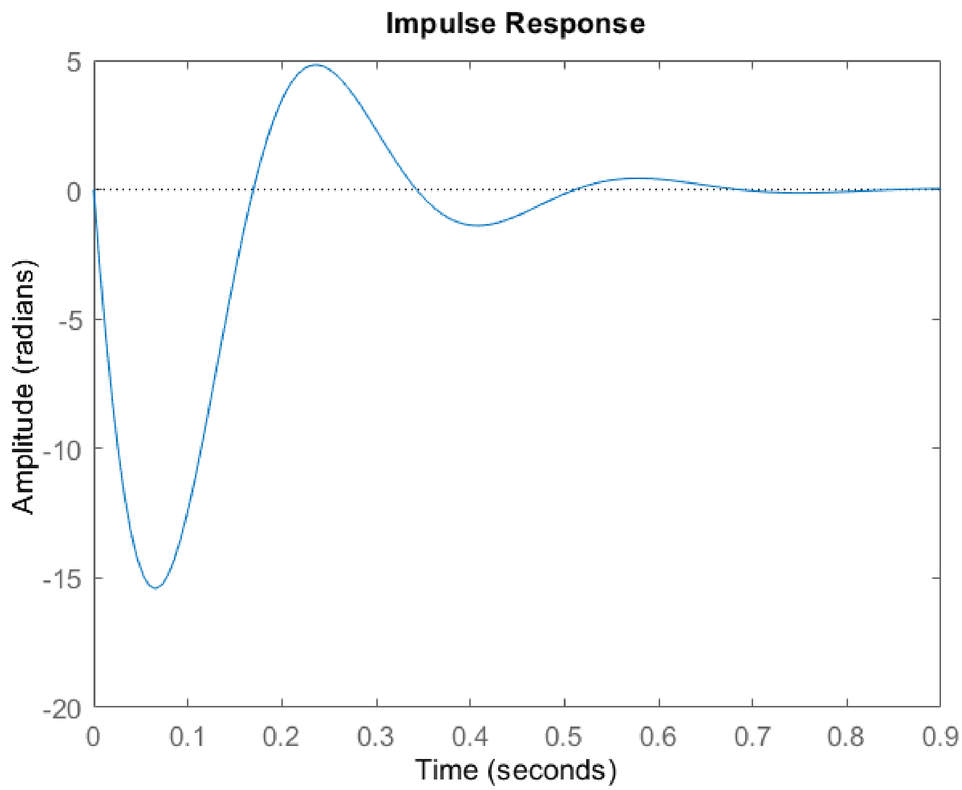

Figure 9: Impulse response of the initial system.

The initial system responded to disturbances by moving in such a manner that the angle was corrected to 0. In order to do this, the initial system's velocity wandered some from 0, and the position wandered significantly. The poles we used were: $-7 \pm 18.35i$ and -5.5. We chose these poles because they have several desired characteristics:

- The complex pole is to the left of the real pole, so the real pole is dominant and the complex pole decays faster. This is desirable because we don't want the oscillations to be dominant.
- The radius of the poles from the axis is relatively large to correspond to a fast rise time.
- The real pole is far from the real axis, so the settling time is low.
- These poles both have negative real components, so the system is stable.

To get from our poles to the control constants, we used the denominator of the transfer function to get a system of equations relating the poles to the control constants.

Using Black's rule, we obtain

$$H_{tot} = \frac{1}{1 - K(s)M(s)H(s)} = \frac{s^3 + \frac{1}{\tau}s^2 - \frac{g}{l}s - \frac{g}{l\tau}}{s^3 + \frac{1}{\tau}s^2 - (\frac{g}{l} - \frac{k_p k_{mot}}{l\tau})s - \frac{g - k_p k_{mot}}{l\tau})}. \tag{8}$$

Therefore, the characteristic polynomial is

$$s^3 + \frac{1}{\tau}s^2 - (\frac{g}{l} - \frac{k_p k_{mot}}{l\tau})s - \frac{g - k_p k_{mot}}{l\tau} \tag{9}$$

A cubic with poles at R, and $\sigma \pm \omega i$ expands into

$$s^3 + (R - 2\sigma)s^2 + (\sigma^2 + 2\sigma R + \omega^2)s - R(\sigma^2 - \omega^2). \tag{10}$$

Using the method of unknown coefficients, we get a system of equations that simplifies to:

$$R = \frac{1}{\tau} + 2\sigma \tag{11}$$

$$k_i = -\frac{(R\tau l(\sigma^2 - \omega^2) - g)}{k_{mot}} \tag{12}$$

$$k_p = -\frac{l\tau(\sigma^2 + 2\sigma R + \omega^2 + \frac{g}{l})}{k_{mot}} \tag{13}$$

Notice that the real pole and the real part of the complex pole are related. This bounds what poles are within the physical constraints of the system (since the link is with the motor time constant). Our chosen poles satisfy this relationship. From these equations, we got $K_i = 3805$ and $K_p = 19983$ using the left wheel motor parameters.

2.2.1 Simulink

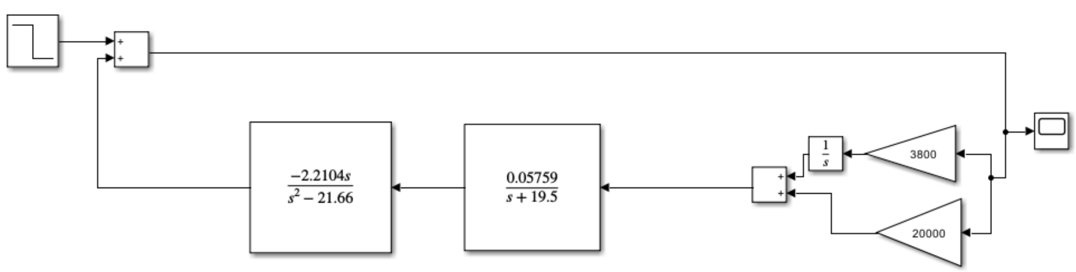

Figure 10: Simulink block diagram.

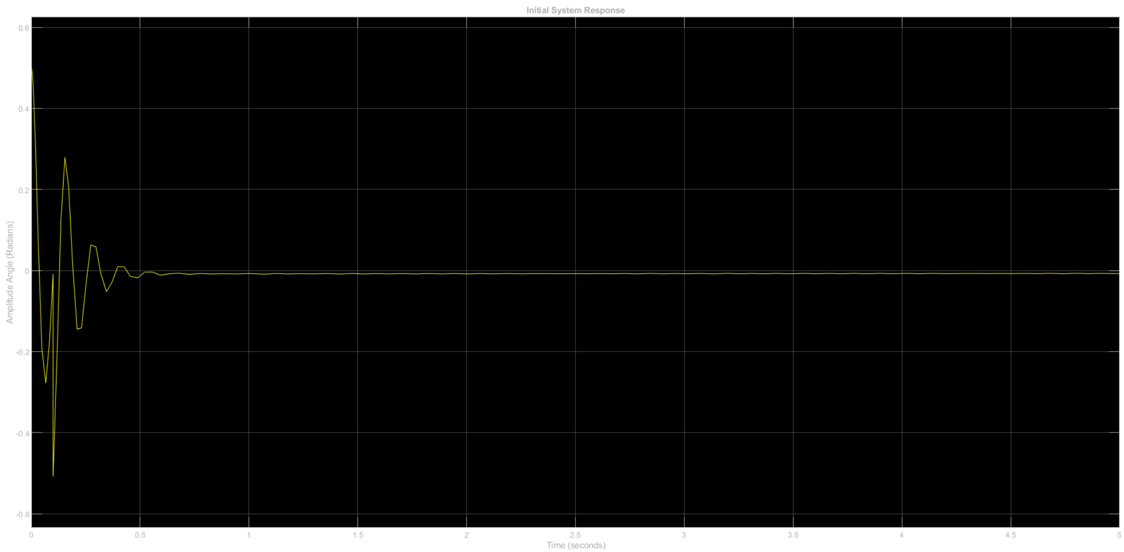

Figure 11: Simulink step response.

3 Stationary System

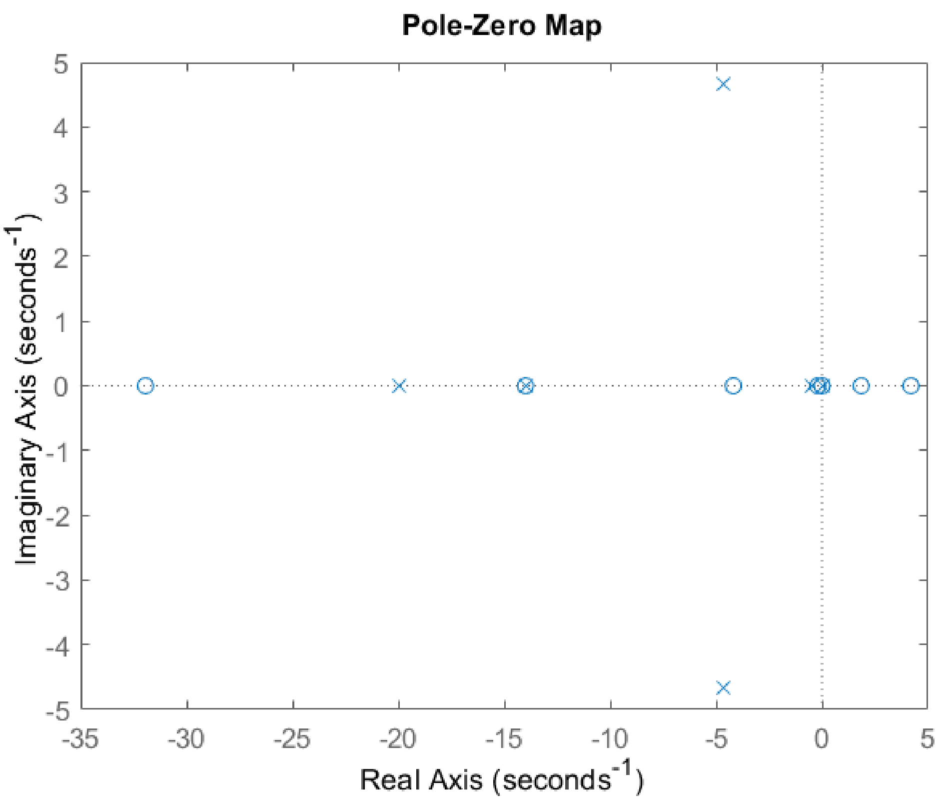

Figure 12: Poles are Xs, zeros are Os.

The stationary system responded to disturbances by moving in such a manner that the angle was corrected to 0. In order to do this, the robot moved forward, crossed the equilibrium position, then tilted back over and moved to its original position with decreasing speed. We wanted the steady state to occur quickly and minimize oscillations. Therefore, we placed two real poles in-between the complex poles and the real axis. Thus, we would have real dominant poles which result in a system with minimal oscillations. This resulted in the double pole at -.5. Finally, we found that moving the final pole far to the left resulted in desired behavior through experimentation. The further the pole was to the left the less reaction there was. This yielded the final pole to be at -20. Thus, the poles in the stationary system were: $\omega_n \pm \omega_n j.$, -.5, -.5, and -20. We chose these poles so that they satisfied the following set of criteria, and within these criteria used trial and error:

- All poles have a negative real component so that the system is stable.
- The radius of the poles from the axis is relatively large to correspond to a fast rise time.
- The oscillations are set equal to the natural frequency.

To get from our poles to the control constants, we used the given Matlab script. The values of the control constants obtained were

K_p	K_i	J_i	C_i	J_p
5256.39	26306.76	-1503.54	-352.54	465.95

3.1 Stationary System

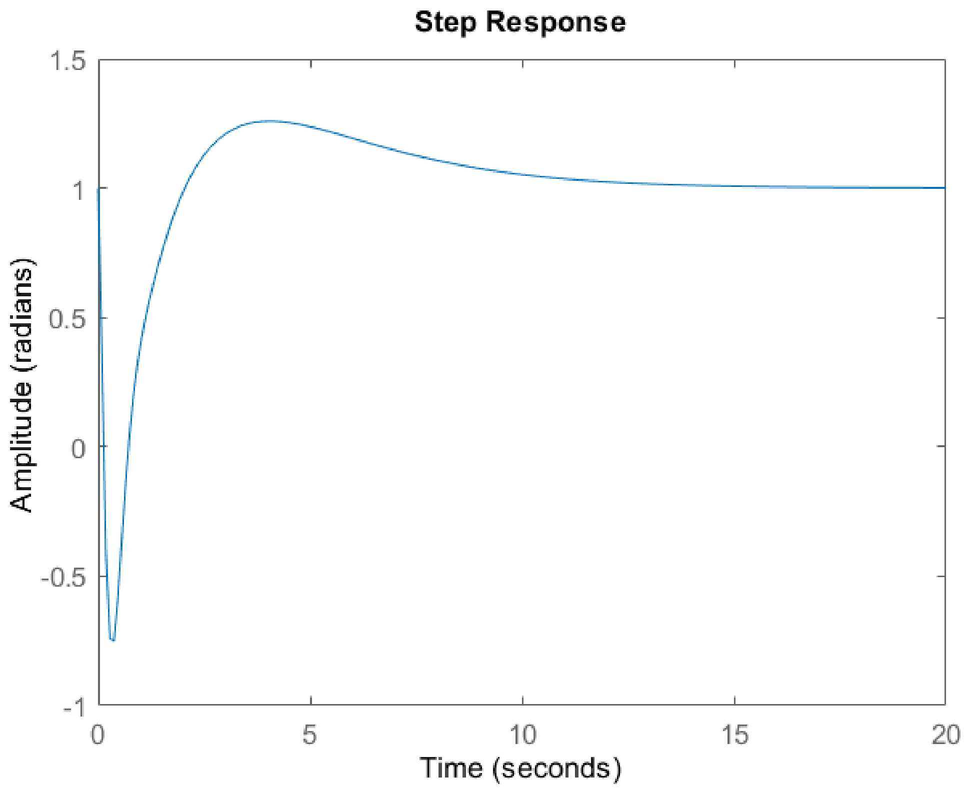

Figure 13: Step response of the stationary system.

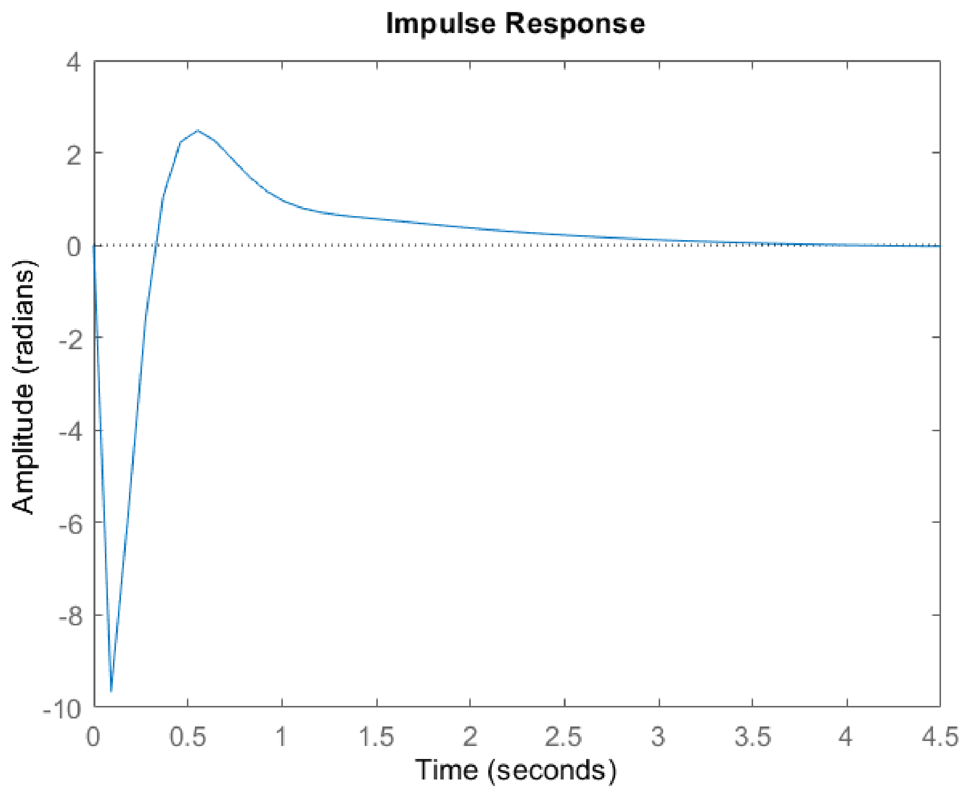

Figure 14: Impulse response of the stationary system.

3.1.1 Simulink

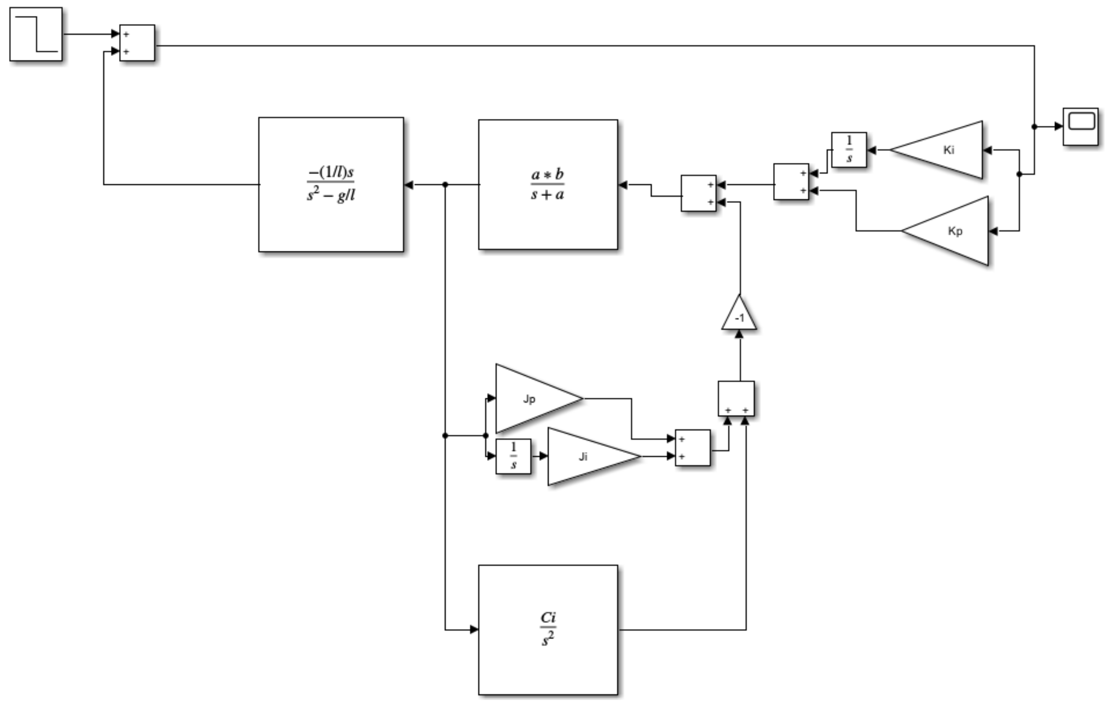

Figure 15: Simulink block diagram of the stationary system.

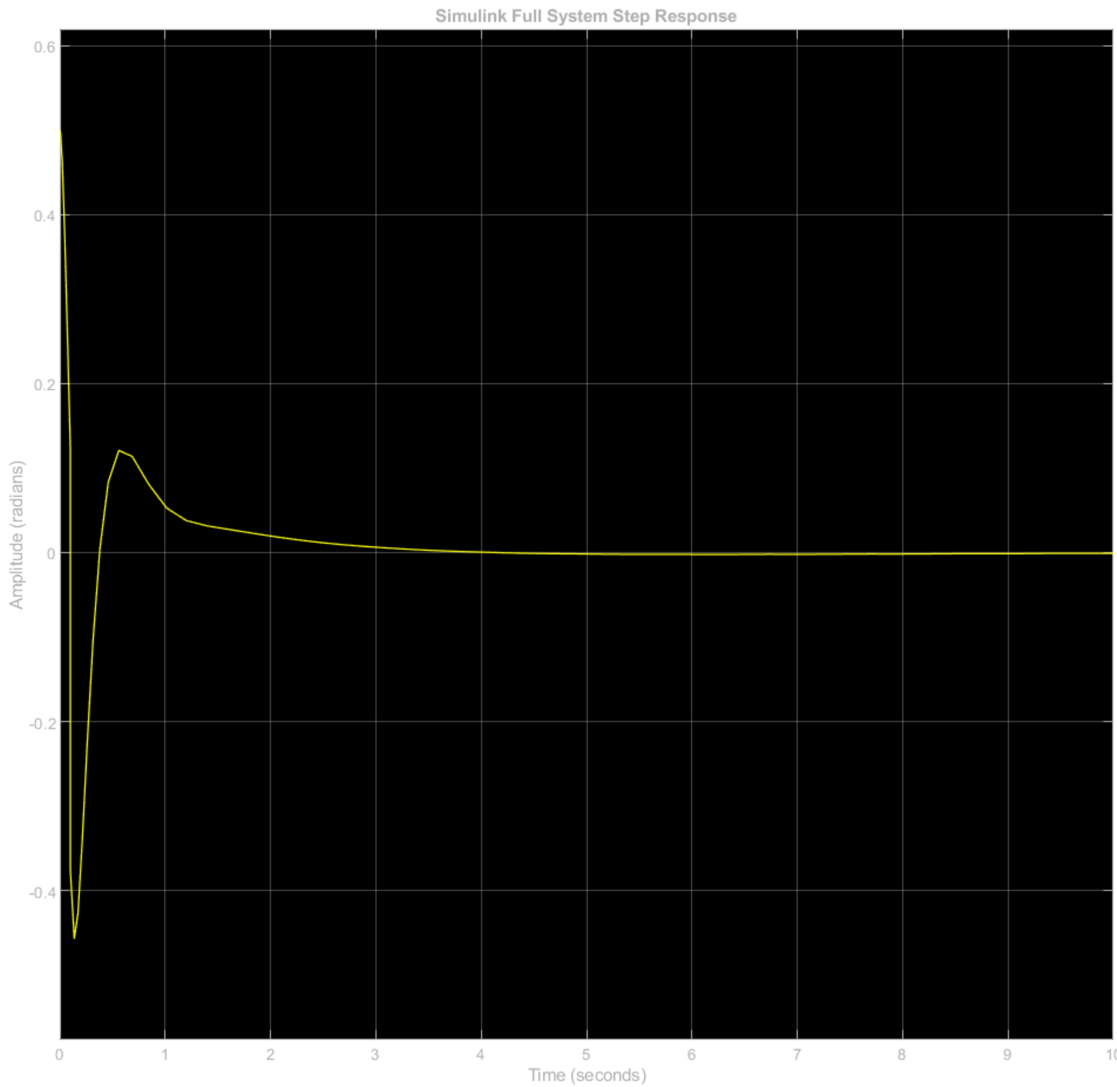

Figure 16: Step response of the stationary system.

4 Code

https://github.com/James-Jagielski/ESA-Rocky-Project

5 Videos

https://youtu.be/_XKrdyzvKyY
https://youtube.com/shorts/JTcX4at39Mg?feature=share